MID SPRING 2026 OF 64

Question 1(a) [3 marks] — CO1

Scenario: A computer graphics lab uses a display unit to show images and animations in multiple colors. The display quality depends on beam alignment and color intensity control. Different internal structures are used in similar versions of this display to improve brightness and color precision. The image must be refreshed repeatedly to remain visible. **Identify** the display device described in the scenario and **compare** its two types.

Answer:

The display device described is the **CRT (Cathode Ray Tube)**, since it requires beam alignment (electron beam control), color intensity control (phosphor excitation), and repeated refreshing to remain visible (raster scan refresh).

There are two types of CRT based on internal structure: **Shadow-Mask CRT** and **Beam-Penetration CRT**. A comparison is shown below.

Feature	Shadow-Mask CRT	Beam-Penetration CRT
Color method	Uses three electron guns (R, G, B) and a metal shadow mask with tiny holes to direct beams to correct phosphor dots	Uses a single electron gun; color depends on how deep the beam penetrates into two layers of phosphor (red inner layer, green outer layer)
Color quality	Produces a wide, realistic range of colors with high resolution	Produces limited colors (usually red, green, and combinations like orange/yellow)
Cost & complexity	More expensive and complex due to three guns and precise alignment	Simpler and cheaper, but lower color accuracy
Common use	Used in high-quality color monitors/TVs	Used in random-scan (vector) displays for simpler color graphics

[বাংলা ব্যাখ্যা]: Scenario তে "beam alignment", "color intensity control", "refreshed repeatedly" — এই keyword গুলো দেখেই বোঝা যায় এটা CRT এর কথা বলছে, কারণ CRT তে electron beam phosphor কে আঘাত করে আলো তৈরি করে এবং সেটা বারবার refresh করতে হয় (otherwise আলো ফিকে হয়ে যায়)। "Different internal structures" বলাতে বোঝা যাচ্ছে দুই ধরনের CRT এর কথা জিজ্ঞেস করা হচ্ছে — Shadow-Mask আর Beam-Penetration।

Question 1(b) [5 marks] — CO1

Question: Explain the concept of the decision parameter in the Mid-Point Circle Drawing Algorithm and describe how the initial and next decision parameters are obtained.

Answer:

Concept of Decision Parameter:

In the Mid-Point Circle Drawing Algorithm, we draw only one octant (1/8th) of the circle and use symmetry to plot the rest. At each step, moving from a known pixel, there are two candidate next pixels — one moving directly right, and one moving diagonally (right and down, since we work in the second octant going from top toward 45°). To decide between them, we evaluate the **circle function**:

$$f_{circle}(x, y) = x^2 + y^2 - r^2$$

- If a point lies **inside** the circle, $f(x,y) < 0$
- If a point lies **on** the circle, $f(x,y) = 0$
- If a point lies **outside** the circle, $f(x,y) > 0$

We test the **midpoint** between the two candidate pixels by substituting it into f(x,y). The sign of this value tells us which candidate pixel is closer to the actual circle boundary — this is the **decision parameter, P_k**.

Initial Decision Parameter:

Starting point is (0, r). The first midpoint to test is (1, r − 1/2). Substituting into the circle equation:

$P_0 = f(1, r - 1/2) = 1^2 + (r - 1/2)^2 - r^2$

$= 1 + r^2 - r - 1/4 - r^2$

$\mathbf{P_0 = 5/4 - r}$

To avoid fractions, this is approximated as:

$\mathbf{P_0 = 1 - r}$

Next (Recurrence) Decision Parameter:

At each step, x increases by 1 ($x_{k+1} = x_k + 1$). Whether y decreases depends on P_k:

- **If $P_k < 0$:** Midpoint is inside the circle → choose pixel (x_k+1, y_k) Next parameter: $\mathbf{P_{k+1} = P_k + 2x_{k+1} + 1}$

- **If P□ ≥ 0:** Midpoint is outside the circle → choose pixel (x□+1, y□−1) Next parameter: **$P_{\square+1}$ = P□ + 2$x_{\square+1}$ + 1 − 2$y_{\square+1}$**

[বাংলা ব্যাখ্যা]: Decision parameter মূলত একটা mathematical test যেটা দিয়ে বোঝা যায় circle এর boundary, দুটো candidate pixel এর মাঝামাঝি বিন্দু (midpoint) এর কোন দিকে আছে — ভেতরে নাকি বাইরে। circle equation এ midpoint বসিয়ে দেখা হয় ফলাফল ধনাত্মক না ঋণাত্মক। এই চিহ্ন (sign) অনুযায়ী পরবর্তী pixel নির্বাচন করা হয় — exactly Bresenham Line Algorithm এর মতোই logic, শুধু এখানে circle equation ব্যবহার হচ্ছে।

Question 2 [6 marks] — CO2

Scenario: A surveillance drone is mapping a restricted rectangular zone using a pixel-based coordinate system. In Figure A, four checkpoints—A, B, C, D—are placed at the corners of the zone. The drone must first trace the bottom boundary of the zone and then draw a direct flight path from the top-left checkpoint to the bottom-right checkpoint using an efficient raster line-drawing technique.

Given coordinates: A(40,50), B(55,50), C(55,40), D(40,40)

(i) Draw the line CD. You can use any suitable algorithm as necessary. (ii) Apply Bresenham's Line Drawing Algorithm to draw the line CA.

Answer (i): Line CD

C(55,40) to D(40,40)

Since y-coordinate is the same for both points (y = 40 for both C and D), this is a **horizontal line**. For a horizontal line, no complex algorithm (like DDA or Bresenham) is needed — we simply keep y constant and decrement x by 1 each step (since we're going from x=55 to x=40).

Pixels plotted: (55,40), (54,40), (53,40), ..., (41,40), (40,40)

This is the simplest case of line drawing — y stays fixed at 40, and x decreases from 55 to 40, one unit at a time.

[বাংলা ব্যাখ্যা]: C ও D দুটোরই y = 40, তাই এটা horizontal line। Horizontal line এ Bresenham এর দরকার নেই — শুধু x একটা একটা করে কমিয়ে (বা বাড়িয়ে) y same রেখে pixel বসালেই হয়ে যায়। "Bottom boundary" বলা হয়েছে কারণ এই zone এর সবচেয়ে নিচের রেখা এটাই (y=40, সবচেয়ে কম y value)।

Answer (ii): Line CA using Bresenham's Algorithm

C(55,40) to A(40,50)

Step 1 — Setup:

$\Delta x = |40 - 55| = 15$ $\Delta y = |50 - 40| = 10$ $m = \Delta y/\Delta x = 10/15 = 0.67$ → **$0 < m \leq 1$, so this is Case 1**

Since x decreases (55 → 40) and y increases (40 → 50):

- x change per step = **−1**
- y change (when it increases) = **+1**

$P_0 = 2\Delta y - \Delta x = 2(10) - 15 = 20 - 15 =$ **5**

Step 2 — Iteration Table:

k	**P_k**	**$P_k \geq 0$?**	**x_{k+1}**	**y_{k+1}**	**P_{k+1}**
0 (start)	5	—	—	—	(55, 40)
1	5	≥0 ✓	55−1=54	40+1=41	5+20−30 = **−5**
2	−5	<0 ✗	54−1=53	41+0=41	−5+20 = **15**
3	15	≥0 ✓	53−1=52	41+1=42	15+20−30 = **5**
4	5	≥0 ✓	52−1=51	42+1=43	5+20−30 = **−5**
5	−5	<0 ✗	51−1=50	43+0=43	−5+20 = **15**
6	15	≥0 ✓	50−1=49	43+1=44	15+20−30 = **5**
7	5	≥0 ✓	49−1=48	44+1=45	5+20−30 = **−5**
8	−5	<0 ✗	48−1=47	45+0=45	−5+20 = **15**
9	15	≥0 ✓	47−1=46	45+1=46	15+20−30 = **5**
10	5	≥0 ✓	46−1=45	46+1=47	5+20−30 = **−5**
11	−5	<0 ✗	45−1=44	47+0=47	−5+20 = **15**
12	15	≥0 ✓	44−1=43	47+1=48	15+20−30 = **5**
13	5	≥0 ✓	43−1=42	48+1=49	5+20−30 = **−5**
14	−5	<0 ✗	42−1=41	49+0=49	−5+20 = **15**
15	15	≥0 ✓	41−1=40	49+1=50	end

Plotted pixels:

(55,40) → (54,41) → (53,41) → (52,42) → (51,43) → (50,43) → (49,44) → (48,45) → (47,45) → (46,46) → (45,47) → (44,47) → (43,48) → (42,49) → (41,49) → (40,50)

[বাংলা ব্যাখ্যা]: এখানে লক্ষ্য করো — x কমছে (55 থেকে 40), তাই প্রতিটা step এ $x_{k+1} = x_k - 1$ (যোগের বদলে বিয়োগ)। y বাড়ছে স্বাভাবিকভাবে (+1) যখন $P_k \geq 0$। Δx (15) ও Δy (10) দিয়ে m=0.67 হয়, যা ≤ 1, তাই এটা Case 1। P_0, recurrence সব একই থাকে — শুধু x এর দিকটা উল্টে গেছে (decreasing) কারণ start point ডানদিকে, end point বামদিকে।

Question 3 [7 marks] — CO2

Scenario: Suppose Mr. Faruk has been appointed to design a video game, where he needs to draw a fan similar to Fig. A. The fan has an outer circular boundary with a diameter of 12 and is centered at (−3, 4). **Apply** the Midpoint Circle Algorithm to calculate the pixel points of the fan's circular boundary, and finally draw a tentative plot of the fan.

Answer:

Step 1 — Setup:

Diameter = 12 → Radius r = 12/2 = **6** Center = (−3, 4)

We first compute the circle assuming center at origin (0,0), then shift all points by adding (−3, 4) at the end.

Step 2 — Initial Decision Parameter:

$P_0 = 1 - r = 1 - 6 = \mathbf{-5}$

Starting point: $(x_0, y_0) = (0, 6)$

Step 3 — Iteration Table (centered at origin):

k	P_k	$P_k < 0$?	(x_{k+1}, y_{k+1})	Next P_{k+1}
0 (start)	−5	—	(0, 6)	—
1	−5	<0 ✓	(1, 6)	−5+2(1)+1 = **−2**
2	−2	<0 ✓	(2, 6)	−2+2(2)+1 = **3**
3	3	≥0 ✗	(3, 5)	3+2(3)+1−2(5) = **0**
4	0	≥0 ✗	(4, 4)	0+2(4)+1−2(4) = **1**
5	1	≥0 ✗	(5, 3)	stop (x ≥ y)

We stop when x ≥ y, since beyond this point symmetry handles the rest.

Step 4 — Points generated in first octant (center at origin):

(0,6), (1,6), (2,6), (3,5), (4,4)

Step 5 — Apply 8-way symmetry (circle centered at origin):

For each point (x,y), the 8 symmetric points are: (x,y), (−x,y), (x,−y), (−x,−y), (y,x), (−y,x), (y,−x), (−y,−x)

This generates all boundary points around the full circle, centered at origin.

Step 6 — Translate to actual center (−3, 4):

Add (−3, 4) to every (x, y) pair obtained above to get the final pixel positions of the fan's circular boundary.

For example: (0,6) → (0−3, 6+4) = (−3, 10) (6,0) → (6−3, 0+4) = (3, 4) (0,−6) → (0−3, −6+4) = (−3, −2) (−6,0) → (−6−3, 0+4) = (−9, 4)

(All other points are translated the same way: x_final = x − 3, y_final = y + 4)

Step 7 — Tentative Plot:

The circular boundary, once all points are translated, forms a circle of radius 6 centered at (−3, 4). The fan blades (as shown in Fig. B) are drawn inside this circular boundary — the circle itself acts as the outer limit within which the curved fan-blade shapes are sketched, similar to the spiral pattern shown in the figure.

[বাংলা ব্যাখ্যা]: প্রথমে radius বের করতে হয় diameter থেকে (r = d/2 = 6)। তারপর Midpoint Circle Algorithm সবসময় ধরে নেয় circle এর center (0,0) এ আছে — কারণ এতে calculation সহজ হয়। তাই প্রথমে origin centered circle এর points বের করি, P_0 = 1−r দিয়ে শুরু করে, প্রতি step এ x বাড়ে, আর P□ এর sign দেখে y কমবে কিনা ঠিক হয়। প্রথম octant এর points পেলে, বাকি circle টা **8-way symmetry** দিয়ে পাওয়া যায় (geometric reflection)। সবশেষে যেহেতু আসল circle টা (−3,4) তে আছে, origin থেকে পাওয়া প্রতিটা point এ (−3, +4) যোগ করে আসল position এ shift করতে হয়। এটাই "fan" এর বাইরের boundary, যার ভেতরে blade গুলো আঁকা হবে figure এর মতো।

Question 4 [4 marks] — CO2

Scenario: The polygonal object ABCDEFG shown in Fig. C represents a 2D object drawn on the Cartesian coordinate system.

a) Translate the object 3 units to the right and 2 units upward. Write the translation matrix and determine the new coordinates of all vertices after translation.

b) Scale the translated object obtained in part (a) using: Sx = 2, Sy = 0.5. Write the scaling matrix and compute the final coordinates of the object.

Reading coordinates from Fig. C:

Based on the figure (approximate grid positions): A(2,3), B(2,1), C(-1,-1), D(-1,-3), E(-5,1), F(-4,1), G(-2,4)

(Note: exact coordinates should be read carefully from your own grid in the question paper — these are estimated from the figure. Use the same method below regardless of exact values.)

Answer (a): Translation

Translation matrix (homogeneous form):

$$T = \begin{bmatrix} 1 & 0 & t_x \ 0 & 1 & t_y \ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 3 \ 0 & 1 & 2 \ 0 & 0 & 1 \end{bmatrix}$$

Formula: x' = x + tx, y' = y + ty where tx = 3, ty = 2

Vertex	Original (x, y)	New x' = x+3	New y' = y+2	Translated (x', y')
A	(2, 3)	5	5	(5, 5)
B	(2, 1)	5	3	(5, 3)
C	(−1, −1)	2	1	(2, 1)
D	(−1, −3)	2	−1	(2, −1)
E	(−5, 1)	−2	3	(−2, 3)
F	(−4, 1)	−1	3	(−1, 3)
G	(−2, 4)	1	6	(1, 6)

[বাংলা ব্যাখ্যা]: Translation মানে পুরো object টাকে একই দূরত্ব সরিয়ে নেওয়া, shape না বদলিয়ে। "3 units right" মানে x তে +3 যোগ হবে, "2 units upward" মানে y তে +2 যোগ হবে। প্রতিটা vertex এ একই rule apply হয়।

Answer (b): Scaling (applied on translated coordinates from part a)

Scaling matrix:

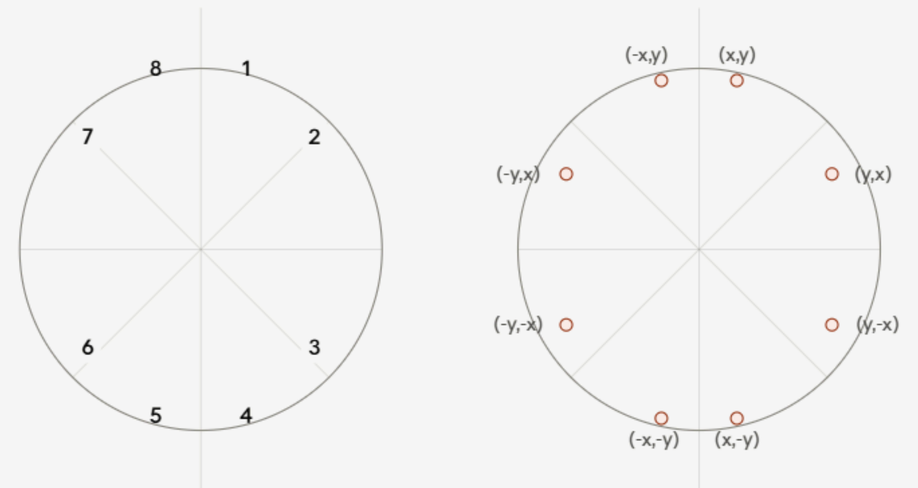
$$S = \begin{bmatrix} S_x & 0 & 0 \ 0 & S_y & 0 \ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \ 0 & 0.5 & 0 \ 0 & 0 & 1 \end{bmatrix}$$

Formula: $x'' = x' \times Sx$, $y'' = y' \times Sy$ where $Sx = 2$, $Sy = 0.5$

Vertex	Translated (x', y')	New x'' = x'×2	New y'' = y'×0.5	Final (x'', y'')
A	(5, 5)	10	2.5	(10, 2.5)
B	(5, 3)	10	1.5	(10, 1.5)
C	(2, 1)	4	0.5	(4, 0.5)
D	(2, −1)	4	−0.5	(4, −0.5)
E	(−2, 3)	−4	1.5	(−4, 1.5)
F	(−1, 3)	−2	1.5	(−2, 1.5)
G	(1, 6)	2	3	(2, 3)

[বাংলা ব্যাখ্যা]: Scaling মানে object কে বড় বা ছোট করা — এখানে Sx=2 মানে x-axis বরাবর দ্বিগুণ চওড়া হবে, আর Sy=0.5 মানে y-axis বরাবর অর্ধেক উচ্চতা হবে (কিছুটা চ্যাপ্টা হয়ে যাবে object টা)। গুরুত্বপূর্ণ বিষয়: scaling সবসময় **translated coordinates** এর উপর **apply** করতে হবে, original coordinates এর উপর না — কারণ question এ বলা আছে "scale the translated object"।

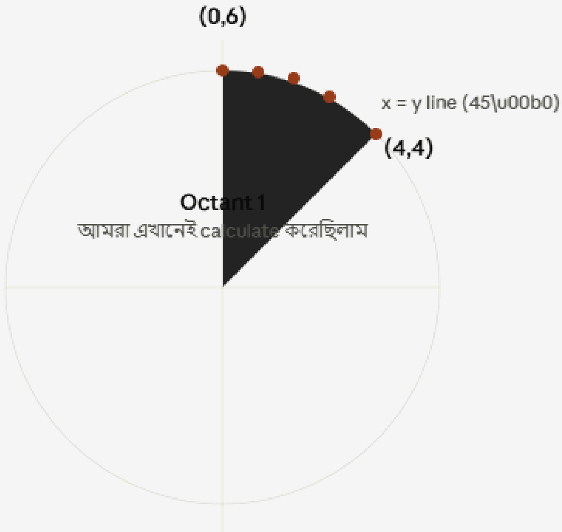
⚠️ গুরুত্বপূর্ণ নোট **Question 4 নিয়ে:** তোমার uploaded paper এর Fig. C এর grid coordinates আমি ছবি থেকে আন্দাজ করে নিয়েছি, কারণ exact pixel positions স্পষ্ট দেখা যাচ্ছে না। পরীক্ষার আগে অবশ্যই তোমার বইয়ের আসল **figure** থেকে **A, B, C, D, E, F, G** এর সঠিক **coordinates verify** করে নিও — পদ্ধতি (method) ঠিক থাকলেও সংখ্যা ভুল হলে নম্বর কাটা যাবে। বাকি সব method (translation formula, scaling formula, matrix) ঠিক আছে এবং exam এ এভাবেই লিখবে।



Octant numbering (1\u20138, clockwise from top)   8-way symmetry: each computed (x,y) gives 8 points

আমরা কোন Octant এর জন্য বের করেছিলাম?

আমরা যে algorithm ব্যবহার করেছিলাম ($P_0 = 1-r$ দিয়ে শুরু, x বাড়ে, P_k এর sign দেখে y কমে কি না) — এটা সবসময় **(0, r) থেকে শুরু করে x=y পর্যন্ত** calculate করে। এই অংশটাকে সাধারণত **Octant 1** (বা বইভেদে Octant 2) বলা হয় — circle এর top থেকে 45° পর্যন্ত অংশ, যেখানে x ছোট থাকে আর y বড় থাকে ($x \leq y$)।



5টি point: (0,6) (1,6) (2,6) (3,5) (4,4) — শুধু এই অংশে